

Programming Language Landscape 2024

Usage Trends, Developer Satisfaction, and Strategic Hiring Implications

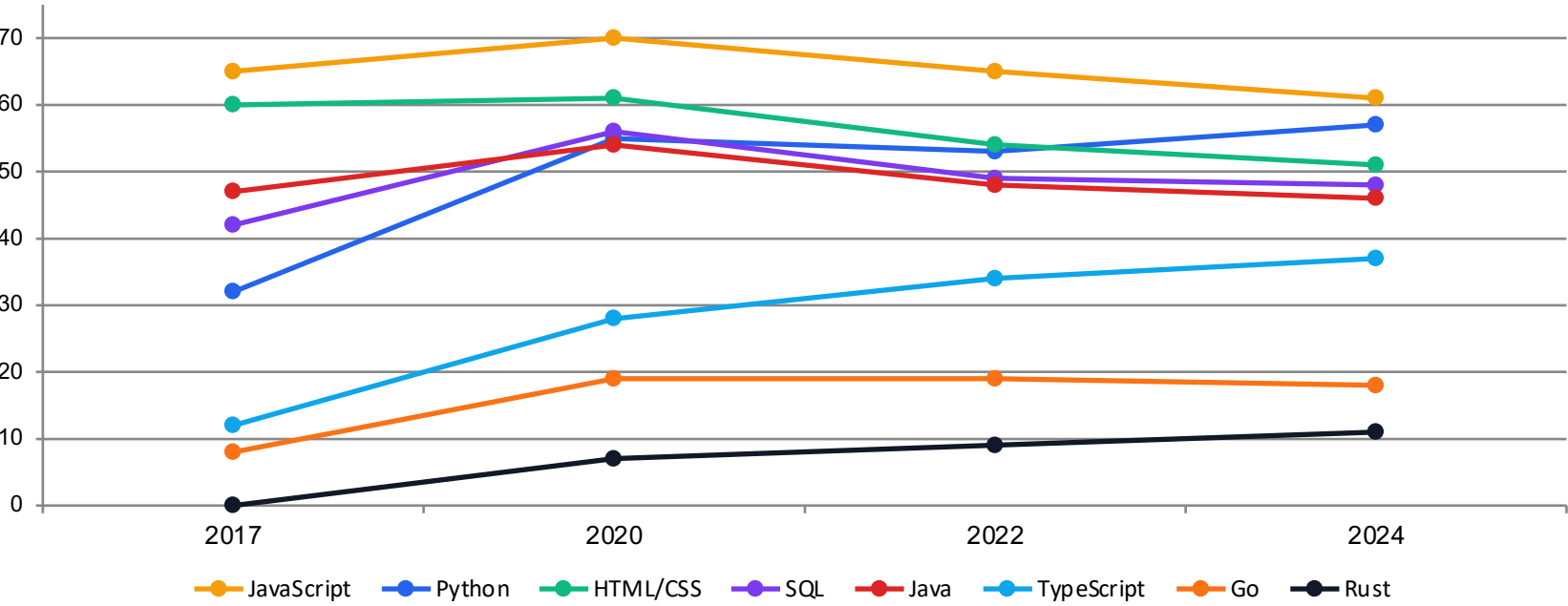
Audience: Engineering Leaders, CTOs, and Developer Experience Teams

Source: JetBrains State of Developer Ecosystem 2024 (n = 23,262)

2025 Strategy Briefing

How Programming Language Usage Has Shifted (2017-2024)

Top languages by usage share (%) across 8 years



Signal: JavaScript remains #1 in overall use (61%), but Python is closing fast (57%) and TypeScript has tripled since 2017.

Primary Language vs. Intent to Adopt: Strong Ecosystem Lock-In

Developers mostly stay within their established stack

Primary Language Share

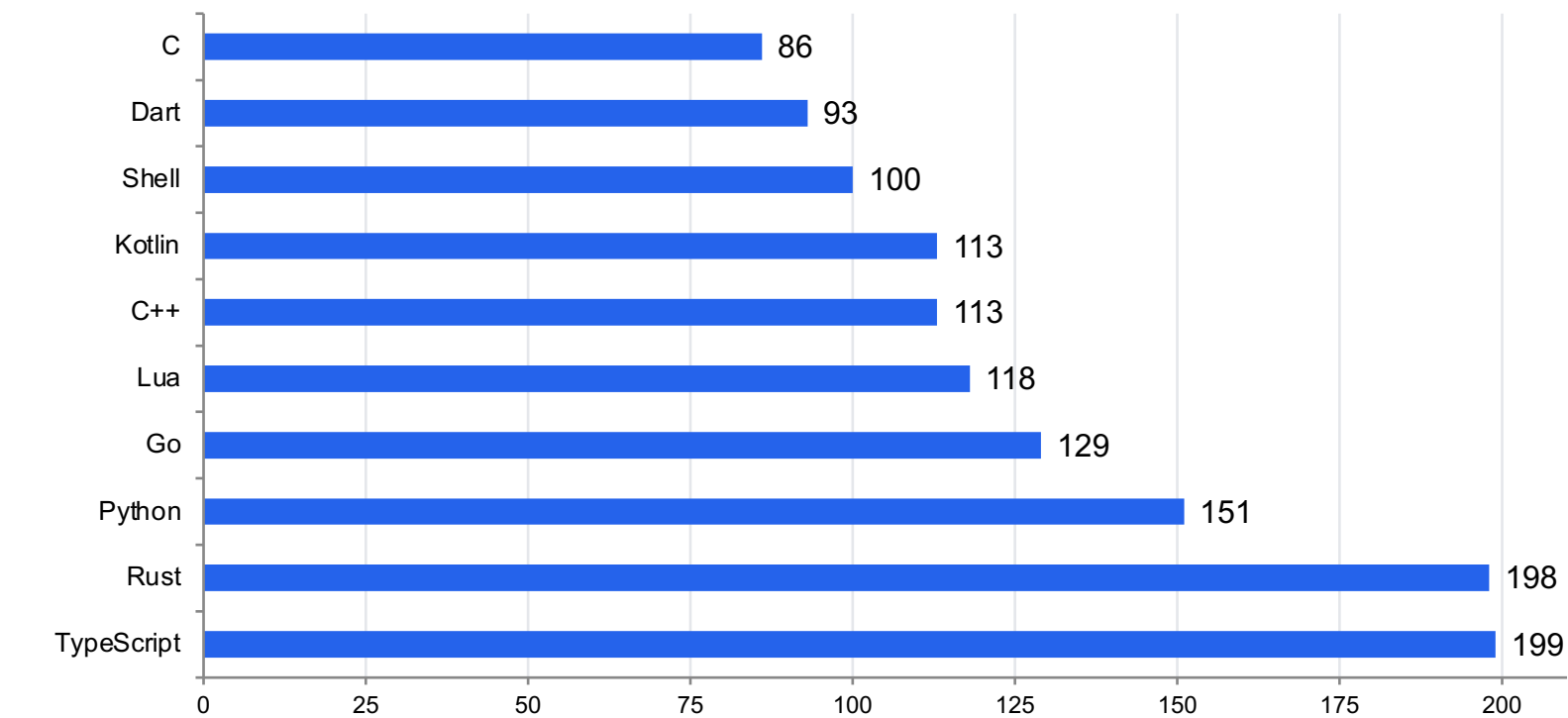
Python 35%
Java 30%
JavaScript 28%
TypeScript 19%
SQL 17%

Planning to Adopt as Primary

Python 6%
TypeScript 5%
Java 4%
SQL 4%

Implication: Platform, tooling, and internal standards influence language inertia more than short-term hype.

Most Loved Languages: Satisfaction Signals Retention Risk/Advantage



TypeScript (199) and Rust (198) top the ranking, indicating strong affinity for modern type safety and tooling.

Language-Domain Fit: One Stack Does Not Win Everywhere

Top language by application domain (from survey data)

AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs & Services	Python	38%

What This Means for Your Technology Strategy in 2025

- 1 Invest in Python for AI/ML and data-heavy initiatives; it leads AI/ML (31%) and data science use cases.
- 2 Standardize TypeScript for frontend/full-stack paths where developer satisfaction and UI dominance matter.
- 3 Prioritize Rust and Go for infrastructure and systems bets where reliability and performance are critical.
- 4 Plan for shrinking PHP and Ruby talent pools as long-term hiring risk in legacy stacks.
- 5 Use developer-experience factors (type safety, tooling quality) as retention levers, not just productivity levers.

Action: Align platform standards, hiring plans, and developer tooling strategy to domain-specific language strengths.